

Data Analytics (CMP330)

Assessment 2

This assessment contributes 50% of the marks for the module.

Date: To be completed during a two-hour lab session on w/c 4 of May. Further details to follow.

Submission: Completion of test on Blackboard and submission of MS Word essay document at the end of the test. Figures, text and references required for part 3 (see below) must also be displayed. These should be included at the end of your document with a clear heading in markdown "Part 3. Presentation of Specified Results".

Feedback date: Feedback will be provided within 20 working days of submission in line with University Guidance.

Overview

You should work on the pre-assignment as specified below in advance of the two-hour lab session where you will be required to use your solutions to the pre-assignment, making minor modifications where necessary, to answer specific questions and finalise your presentation of results. The deliverables of the task will be submitted to Blackboard by the end of the lab. This will include answers to the questions in a Blackboard test and the submission of your R Notebook. During the lab session, you may use your own notes and resources on Blackboard but may not use other online resources. Note that lab machines must be used during the lab session, so it is important to make sure your code runs on these machines in advance. Bear in mind that you may need to install multiple packages.

Pre-Assignment

The dataset to be used for this assessment is the car.csv file that you can access in the Assessment and Feedback area in Blackboard. You can find out details about the original Car Evaluation dataset from the UC Irvine Machine Learning Repository:

<https://archive.ics.uci.edu/dataset/19/car+evaluation>

However, please note that you should not download the data from this repository but should instead download the slightly modified version from Blackboard. Once downloaded you should import it using the following code:

```
df = read.csv("car.csv")
df = data.frame(lapply(df, factor))
```

```
df = read.csv("car.csv")

df = data.frame(lapply(df, factor))

head(df)

str(df)
```

Your task is to apply machine learning (ML) algorithms to classify cars in terms of their acceptability ("unacc", "acc", "good", "vgood") based on the other features in the dataset. Your analysis should include work as described below:

1. Exploratory data analysis and data preparation (20%)

- i) You should be able to determine the number of cars that have a specified value of a given variable (e.g. the number with acceptability = "unacc", "acc", etc.) and also the number of cars that have a specified value of one variable and a specified variable of another variable (e.g. the number of websites for which acceptability = "unacc" and another variable = "low"). (You can use the table function.) ii) You should be able to plot bar charts to show the distribution for a given variable on its own and also for a given variable with a breakdown by values of another variable (see the example of survival broken down by male/female on slide 58 of the week 3 lecture slides – you should use the ggplot2 package for these plots).

```
table(df$acceptability)

table(df$buying, df$acceptability)

summary(df)
```

```
library(ggplot2)

df$acceptability = factor(df$acceptability, levels =
  c("unacc", "acc", "good", "vgood"))

ggplot(df, aes(x=acceptability)) + geom_bar()

ggplot(df, aes(x=acceptability, fill=buying)) + geom_bar()
```

- ii) Since the variables are not numerical, you should find the mutual information between variables rather than the correlation. To do this you should install the `infotheo` package and then call the function `mutinformation` in the same way you would call the correlation function `cor` (i.e. for two variables or for a dataframe).

```
install.packages("infotheo")

library("infotheo")
m = mutinformation(df) m

# Could use following code for each row, replacing [-1,1] with [-2,2], etc.
max(m[-1,1])
```

- iii) You should be able to use the chi-squared test for independence between two variables (see week 5 slides).

```
chisq.test(df$buying,df$acceptability)

chisq.test(df$buying,df$maint)
```

- iv) You should use the following code to split the data into training and testing sets (replacing 'df' with the name you have used for the dataframe):

```
library(caTools)
set.seed(123)
split = sample.split(df$Result, SplitRatio = 0.8)
df_train = subset(df, split == TRUE)
df_test = subset(df, split == FALSE)
```

```

library(caTools)

set.seed(123)

split = sample.split(df$acceptability, SplitRatio = 0.8)

df_train = subset(df, split == TRUE)

df_test = subset(df, split == FALSE)

summary(df_train$acceptability)

summary(df_test$acceptability)

prop.table(table(df_train$acceptability))

prop.table(table(df_test$acceptability))

```

2a. Data Analytics solution, part A (40%): Application of machine learning

- i) You should implement a basic version of logistic regression (this will need to be the multinomial version since there are three classes), decision trees (using the C5.0 algorithm), naïve Bayes, SVM and neural networks using the functions as illustrated in the document Guidance for Assessment 2 in the Assessment and Feedback area of Blackboard. After training each algorithm on the training set, you should then evaluate them on the test set using appropriate evaluation metrics.

Decision Tree, C5.0

```

library(caret)
library(C50)
dt_model <- C5.0(df_train[-7], df_train$acceptability)
dt_pred <- predict(dt_model, df_test)
confusionMatrix(dt_pred, df_test$acceptability)

```

Naive Bayes

```

library(e1071)
set.seed(123)
nb_model = naiveBayes(df_train[-7], df_train$acceptability)
nb_pred = predict(nb_model, df_test) confusionMatrix(nb_pred,
df_test$acceptability)

```

Logistic Regression

```
library(nnet)
lg_model <- multinom(acceptability ~ ., data = df_train)
lg_pred <- predict(lg_model, df_test)
confusionMatrix(lg_pred, df_test$acceptability)
```

SVM

```
library(kernlab)
svm_model <- ksvm(acceptability ~ ., data = df_train, kernel = "vanilladot")
svm_pred <- predict(svm_model, df_test)
confusionMatrix(svm_pred, df_test$acceptability)
```

SVM

```
library(e1071)
svm_model <- svm(acceptability ~ ., data = df_train)
svm_pred <- predict(svm_model, df_test)
confusionMatrix(svm_pred, df_test$acceptability)
```

Neural Network

see <https://www.geeksforgeeks.org/r-language/neural-networks-using-the-r-nnet-package/>

```
library(nnet)
nn_model <- nnet(acceptability ~ ., data = df_train, size = 5)
nn_pred <- predict(nn_model, df_test, type="class")
confusionMatrix(factor(nn_pred), df_test$acceptability)
```

- ii) You should use the train function from the caret package (as we did for naïve Bayes in the week 10 lab – see also the Guidance for Assessment 2 document) to tune hyperparameters and make use of cross-validation in the training process. The resulting model should then be evaluated on the test data.

#Decision Tree with CV

```
ctrl = trainControl(method='cv',number=10)
set.seed(123)
dt_cv_model = train(df_train[,-7],df_train[,7],'C5.0',trControl=ctrl)
dt_cv_model
dt_cv_pred = predict(dt_cv_model,df_test)
confusionMatrix(dt_cv_pred, df_test$acceptability)
```

Naive Bayes with CV

```
ctrl = trainControl(method='cv',number=10)
set.seed(123)
nb_cv_model = train(df_train[,-7],df_train[,7],'nb',trControl=ctrl)
nb_cv_model
nb_cv_pred = predict(nb_cv_model,df_test)
confusionMatrix(nb_cv_pred, df_test$acceptability)
```

Logistic Regression with CV

```
ctrl = trainControl(method='cv',number=10)
set.seed(123)
lg_cv_model = train(df_train[,-7],df_train[,7],'multinom',trControl=ctrl,
  trace=FALSE)
lg_cv_model
lg_cv_pred = predict(lg_cv_model,df_test)
confusionMatrix(lg_cv_pred, df_test$acceptability)
```

Neural Network with CV

```
ctrl = trainControl(method='cv',number=10)
set.seed(123)
nn_cv_model = train(df_train[,-7],df_train[,7],'nnet',trControl=ctrl,trace=FALSE)
nn_cv_model
nn_cv_pred = predict(nn_cv_model,df_test)
confusionMatrix(nn_cv_pred, df_test$acceptability)
```

SVM with CV

```
ctrl = trainControl(method='cv',number=10)
set.seed(123)
svm_cv_model <- ksvm(acceptability ~ ., data = df_train, kernel = "vanilladot",
  cross=10)
svm_cv_model
svm_cv_pred <- predict(svm_cv_model, df_test)
confusionMatrix(svm_cv_pred, df_test$acceptability)
```

- iii) You should apply a filter-based approach to feature selection by finding the mutual information (see 1(iii)) between each feature and the target variable in the training data. Select the x features with the highest value and then re-run the basic versions of the

algorithms as in 2a(i) above using only these features (you can try this for a range of values of x).

Feature Selection

```
m = mutinformation(df_train)
m
m[7,-7]
```

Decision Tree with Feature Selection

```
dt_fs_model <- C5.0(df_train[,4:6], df_train$acceptability)
dt_fs_pred <- predict(dt_fs_model, df_test)
confusionMatrix(dt_fs_pred, df_test$acceptability)
```

Naive Bayes with Feature Selection

```
nb_fs_model = naiveBayes(df_train[,4:6],df_train$acceptability)
nb_fs_pred = predict(nb_fs_model,df_test)
confusionMatrix(nb_fs_pred, df_test$acceptability)
```

Logistic Regression with Feature Selection

```
lg_fs_model <- multinom(acceptability ~ persons + lug_boot + safety, data =
  df_train)
lg_fs_pred <- predict(lg_fs_model, df_test)
confusionMatrix(lg_fs_pred, df_test$acceptability)
```

2b. Data Analytics solution, part B (10%): Improving model performance

You should carry out further tuning of the hyperparameters on the training set to see how the accuracy varies as the hyperparameters change as we did for the trial's hyperparameter for the C5.0 approach in the week 9 lab, e.g. trials for C5.0, fL for naïve Bayes, decay for neural networks. See Guidance for Assessment 2.

Tuning for Neural Network

```
ctrl = trainControl(method="repeatedcv",number=10,repeats=10)
grid <- expand.grid(size=5,decay=c(0,0.001,0.005,0.01,0.02,0.05,0.1))
set.seed(123)
m <- train(df_train[,-7],df_train[,7], method = "nnet", trControl = ctrl,tuneGrid =
  grid,trace=FALSE)
plot(m)
```

Tuning for DT

```
ctrl = trainControl(method="repeatedcv",number=10,repeats=10)
grid <- expand.grid (model="tree",trials = c(1,5,10,15,20,25,30,40,50,60,80,100),
  winnow=FALSE)
set.seed(123)
m <- train(df_train[,-7],df_train[,7], method = "C5.0", trControl = ctrl,tuneGrid =
  grid)
plot(m)
```

3. Presentation of results for above sections (30%)

As part of your submission during the lab session, you will be asked to upload your R Notebook which should include figures of results (e.g. for 1(ii), 2a(iii) and 2b) with a brief description, Details of exactly which figures must be included will be specified during the final assessment in the lab, but you should generate relevant figures in advance, so that it will be easy to make adjustments on the day. You should also provide references to sources that you have used in your work. This should be done using the Harvard referencing system.

Plagiarism

When submitting your work, you are agreeing to the following statement:

“I declare that this is all my own work. Any material I have referred to has been accurately referenced and any contribution of Artificial Intelligence technology has been fully acknowledged. I understand the importance of academic integrity and have read and understood the University’s General Regulation: Student Academic Integrity and the Academic Misconduct Procedure. I understand that I must not upload my work before, during or after submission to any unapproved plagiarism detectors or answer sharing platforms, or equivalent, and that only University-approved platforms should be used.”

University policies on academic integrity can be found at

<https://www.ulster.ac.uk/learningenhancement/resources/academic-integrity-and-plagiarism>

Rubric for Assessment

Criteria (100%)	0-39% (Fail)	40-49% (3 rd)	50-59% (2.2)	60-69% (2.1)	70-79% (1 st)	80-100% (High 1 st)
Exploratory data analysis and data preparation 20%	Very basic EDA and little or no data preparation.	Adequate EDA with some data Preparation.	Good EDA with reasonably detailed data preparation.	Very good EDA with thorough data preparation.	Excellent EDA with detailed and thorough data preparation.	Outstanding EDA extensive and thorough data preparation.
Data Analytics Solution 50%	Very limited solution with very few correct results and little or no evaluation.	Adequate solution with some correct results and basic evaluation.	Good solution with good range of correct results and clear evaluation.	Very good solution with wide range of correct results and thorough evaluation.	Excellent solution with very wide range of correct results and extensive evaluation.	Outstanding solution with excellent range of correct results involving advanced techniques with extensive evaluation.
Presentation of Results 30%	A very poor presentation of results that lacks detail or proper formatting. No references.	Adequate presentation of results but with limited detail or proper formatting. Few if any references.	A good presentation of results with reasonable level of detail and appropriate formatting. Some references included.	A very good presentation of results with appropriate level of detail and good formatting. Some relevant references cited properly.	Excellent presentation of results with high level of detail and excellent formatting. Good use of relevant references cited properly.	An outstanding presentation of results with extensive level of detail and excellent formatting. Excellent use of highly relevant references cited properly.

